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Review text:

The degree/diameter problem asks to find the maximum order of a graph with maximum degree d and diameter k . The *Moore bound* refers to a natural upper bound for the order of such a graph, and graphs that attain this bound are called *Moore graphs*. An important open problem is whether there exists a Moore graph of diameter $k = 2$ and maximum degree $d = 57$. For a dynamic survey of the area, see Miller–Širáň [Electron. J. Combin. **DS14** (2005), 61 pp.; MR4336216].

The degree/diameter problem has also been studied for directed graphs with maximum outdegree d and diameter k , for which the Moore bound is $M_{d,k} = 1 + d + d^2 + \cdots + d^k$. It turns out that Moore digraphs do not exist unless $d = 1$ or $k = 1$. Then, a natural problem is to investigate *almost Moore digraphs*, also known as (d, k) -*digraphs of defect 1*, which are digraphs of maximum outdegree d and diameter k whose order is one less than the Moore bound. Almost Moore digraphs have been shown to exist for $k = 2$; see Fiol–Alegre–Yebra [ACM SIGARCH Comput. Archit. News **11** (1983), 174–177] and Gimbert [Discrete Math. **231** (2001), no. 1–3, 177–190; MR1821958]. It is believed that almost Moore digraphs do not exist for $k \geq 3$, and there are several partial results in the literature that support this conjecture. For instance, Miller–Fris [In: R. S. Rees (Ed.), *Graphs, Matrices, and Designs*, Lecture Notes in Pure and Appl. Math., vol. 139, Marcel Dekker, New York, 1993, 269–278; MR1209196] showed the nonexistence for $d = 2$, Baskoro–Miller–Širáň–Sutton [J. Graph Theory **48** (2005), no. 2, 112–126; MR2110581] showed the nonexistence for $d = 3$, and

Conde–Gimbert–González–Miret–Moreno showed the nonexistence for $k = 3$ [Electron. J. Combin. **15** (2008), no. 1, #R87, 14 pp.; MR2426150] and $k = 4$ [Electron. J. Combin. **20** (2013), no. 1, #P75, 14 pp.; MR3040637].

In this article, the authors show that in order to prove the non-existence of almost Moore digraphs, it suffices to restrict attention to those with the following specific structure. It is known that in any (d, k) -digraph G of defect 1, for each $v \in V(G)$ there is a unique vertex $r(v) \in V(G)$, called the *repeat* of v , such that there are exactly two $v \rightarrow r(v)$ walks in G of length at most k . The map $r: V(G) \rightarrow V(G)$ turns out to be a permutation of $V(G)$ (and also an automorphism of G), as shown by Baskoro–Miller–Plesník–Znám [J. Graph Theory **20** (1995), no. 3, 339–349; MR1355433]. Let m_i be the number of cycles of length i in the cycle decomposition of r as a permutation of $V(G)$, and call the sequence $(m_0, m_1, m_2, \dots)$ the “permutation cycle structure” of G . Then, the authors prove that any almost Moore digraph G with diameter $k > 2$ has an induced subgraph H that is an almost Moore digraph of order N whose permutation cycle structure is $(k, 0, \dots, 0, m_s, 0, \dots)$ where $N = k + sm_s$, or $(0, \dots, 0, m_s, 0, \dots)$ where $N = sm_s$, or $(0, \dots, 0, m_s, 0, \dots, 0, m_{2s}, 0, \dots)$ where $N = sm_s + 2sm_{2s}$.

The subgraph H is induced on the set S of vertices whose orders divide ℓ , where $\ell = s$ or $2s$ as the case may be; here, the *order* of a vertex v is defined to be the least integer $t > 0$ such that $r^t(v) = v$. To prove the results, the authors work with a block decomposition of the adjacency matrix of G with blocks defined by S and $V(G) \setminus S$. The results in this paper also generalize the results of Sillasen [Electron. J. Graph Theory Appl. **3** (2015), no. 1, 1–7; MR3333722] for $d = 4$ and $d = 5$.